

Introduction to Group theory, Lecture notes

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Lesson 1, 23/09

ON TABLET NOTES

Lesson 2, 28/09

1 Again on SU(2)

The matrices of SU(2) can be seen as matrices in the form

$$g = \begin{pmatrix} x_0 + ix_3 & x_2 + ix_1 \\ -x_2 + ix_1 & x_0 - ix_3 \end{pmatrix}, \text{ with } x_0^2 + x_1^2 + x_2^2 + x_3^2 = \det(g) = 1$$

It can be also represented by bigger matrices, but some of them are trivial, like

$$g = \begin{pmatrix} x_0 + ix_3 & x_2 + ix_1 & 0 \\ -x_2 + ix_1 & x_0 - ix_3 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \text{ with } x_0^2 + x_1^2 + x_2^2 + x_3^2 = \det(g) = 1$$

which is the same as before, but with a bunch of zeros.

2 Generators

The generators are defined as the "directions" in which matrices can "move away" from the identity matrix: it is clear that

$$g = x_0 \mathbb{I} + ix_i \sigma^i$$

where σ^i are the Pauli matrices; since this is a continuous group, the infinitesimal movement in the direction σ^1 can be written as

$$g = \mathbb{I} + i\varepsilon \sigma^1 + O(\varepsilon^2).$$

Consider the set of generators of a continuous (Lie) group $\{T^a\}$: the commutator between them is

$$[T^a, T^b] = T^a T^b - T^b T^a \stackrel{\text{Lie}}{=} if_{ab}^c T^c$$

in the case of $SU(2)$, the structure constants turn out to be $f_{ab}^c = \varepsilon_{ab}^c$, which is the Levi-Civita symbol.

The algebra of the generators of $SO(3)$ is exactly the same.

3 Lorentz-Poincaré Group

This is the group of transformations of Special Relativity: using 4-vectors, a kind of transformation that obeys the rules of SR is a boost along the x-axis

$$B_1 = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

this is a kind of rotation in a Minkowski space; regular spacial rotations also obey SR

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & & & \\ 0 & R & & \\ 0 & & & \end{pmatrix}$$

where $R \in SO(3)$. There are two other transformations, the parity and time-inversion:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

This group has 6 generators: 3 rotations from $SO(3)$ and 3 boosts (along the 3 axes); what are the generators for the boosts?

From their definitions,

$$\gamma^2 - (\gamma\beta)^2 = 1$$

so they can be parametrized like

$$\begin{cases} \gamma = \cosh(\eta) = 1 + O(\eta^2) \\ \beta\gamma = \sinh(\eta) = \eta + O(\eta^2) \end{cases}$$

where η is called rapidity.

$$B_1(\eta) = \mathbb{I} + i\eta \begin{pmatrix} 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + O(\eta^2)$$

So, the generators of rotations are $\{J_i\}$, with the aforementioned algebra.

Similarly, the generators of the boosts $\{K_i\}$ have commutation relations

$$[K^a, K^b] = -i\varepsilon^{ab}_c J^c$$

Trying to mix the generators yields

$$[J^a, K^b] = i\varepsilon^{ab}_c K^c.$$

This can be written in a cleaner way, defining

$$N_a = \frac{J_a + iK_a}{2}$$

$$M_a = \frac{J_a - iK_a}{2},$$

it can be proven that they satisfy 2 closed $SU(2)$ algebras:

$$[N^a, N^b] = i\varepsilon^{ab}_c N^c$$

$$[M^a, M^b] = i\varepsilon^{ab}_c M^c$$

$$[N_a, M_b] = 0$$

Since $SU(2)$ is the angular momentum group, an "azimuthal number" can be defined for the two algebras, p, q : depending on the values of these eigenvalues the field satisfying the Lorentz invariance has different properties.

p	q	Name
0	0	Scalar
1/2	0	Spin 1/2 right
0	1/2	Spin 1/2 left

The addition of the Poincaré group are spacetime translations, which are related to the 4-momentum P^μ

4 Internal degrees of freedom

Some particles are invariant under transformations of "internal" degrees of freedom, like colour charge in quarks: if these transformations are local, they are called **gauge transformations**; locally it is always possible to redefine what the colour "blue" is for a quark.

Why is it possible to assume local symmetries? Because it makes the mediator fields come out of the equations (what?).

In the standard model, these internal degrees of freedom are

$$\underbrace{SU(3)}_{\text{QCD}} \times \underbrace{SU(2) \times U(1)}_{\text{Electroweak}}$$

5 Other definitions on groups

Def. A group $(AG4, \cdot)$ is Abelian if and only if

$$\forall g_1, g_2 \in AG4, g_1 \cdot g_2 = g_2 \cdot g_1$$

Observation 1 The definition of group implies the uniqueness of the identity element.

Proof (by contradiction) Consider 2 identity elements

$$e \neq f :$$

then their product must be, at the same time

$$\begin{cases} e \cdot f = f \\ e \cdot f = e \end{cases} .$$

This contradicts the assumption $e \neq f$ $\square$

Observation 2 The inverse of every element is unique. The proof is similar to the previous one

Def. A magma is a set with an internal binary operation.

Def. A magma $(M, \cdot)$ is called a semi-group if and only if the binary operation is associative

Def. A semi-group with an identity element is called a monoid.

$$\text{Magma} \supset \text{Semi-group} \supset \text{Monoid} \supset \text{Group} \supset \text{Abelian Group}$$

The set of a group can be

- Finite;
- infinite and discrete;
- infinite and continuous.

The smallest non-trivial group is called $\mathbb{Z}_2$:

$$\mathbb{Z}_2 = \{e, a\}$$

and is called cyclic group of order 2; in order to specify the algebraic structure, all the possible products must be determined: a useful tool is a Cayley table In this case, the only element that has to be explicitly

	e	a
e	$e \cdot e$	$e \cdot a$
a	$a \cdot e$	$a \cdot a$

specified is $a \cdot a$. This must be $a \cdot a = e$, because supposing $a \cdot a = a$ implies

$$\begin{aligned} a \cdot a &= a \\ (a \cdot a) \cdot a^{-1} &= a \cdot a^{-1} \\ a \cdot (a \cdot a^{-1}) &= e \\ a \cdot e &= e \\ a &= e \end{aligned}$$

which is absurd.

This group can be realized in many ways (see lesson 1) and also with

$$(\{\text{False}, \text{True}\}, \text{XOR})$$

Def. Given two groups $(G_1, \cdot)$, $(G_2, \circ)$, their **direct group product** is the group $G_1 \times G_2$ made from the set of ordered pairs

$$(a_1; a_2), \text{ with } a_i \in G_i,$$

and the internal operation $\square = (\cdot; \circ)$ that works as follows

$$(a_1; a_2) \square (b_1; b_2) := (a_1 \cdot b_1; a_2 \circ b_2)$$

6 Examples of groups

Integers: The set $\mathbb{Z}$ is an Abelian group with the regular sum $+$, identity element 0 and inverse $n^{-1} := -n$.

Reals: The set of real numbers is similarly an Abelian group with the regular sum, but to construct a group with the multiplication the number 0 must be taken out: $(\mathbb{R} \setminus 0, \cdot)$ is an Abelian group, with 1 as the identity and $a^{-1} = \frac{1}{a}$.

$n \times m$ matrices over $\mathbb{R}$: These sets are groups with the matrix sum, as they are equivalent to $n \times m$ copies of $(\mathbb{R}, +)$.

Invertible matrices: These sets are groups with the matrix multiplication and the inverse is the matrix inverse.

Lesson 3, 30/09

7 Other examples

Dihedral groups: The set is

$$D_n = \{\text{Rotations and reflections that preserve the shape of a regular } n\text{-sided polygon}\}$$

and the group product is the "composition" of these functions.

The smallest group of this family, D_3 , has 6 elements: 3 rotations of angles $0, \frac{2\pi}{3}, \frac{4\pi}{3}$ and 3 reflections about the axes of the triangle.

In D_4 , there are 4 rotations, but the number of reflections is different: 2 about diagonal axes (as the other 2 are the same) and 2 about the axes of the sides (as the other 2 are the same).

In general, $\forall n \geq 3$, the group D_n has $2n$ elements, so its order is

$$|D_n| = 2n :$$

n rotations of angles $\frac{2k\pi}{n}$, with $k = 0, 1, \dots, n-1$ and n reflections. No dihedral group is Abelian, since rotations and reflections do not commute.

Linear transformations of orthonormal bases in $\mathbb{R}^2$: There are only two kinds of transformations that preserve the orthonormality

$$M(\alpha) = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix} \quad N(\alpha) = \begin{pmatrix} \cos \alpha & \sin \alpha \\ \sin \alpha & -\cos \alpha \end{pmatrix}, \text{ with } 0 \leq \alpha \leq 2\pi$$

the first has $\det(M) = 1$, while the second has $\det(N) = -1$. The group product is again the matrix multiplication.

This group is called the **2-dimensional orthogonal group over $\mathbb{R}$** $O(2, \mathbb{R})$, and the matrices with $\det(M) = 1$ form a subgroup called **special orthogonal group**

$$SO(2, \mathbb{R}) = \{M \in O(2, \mathbb{R}) | \det(M) = 1\}$$

The special group is Abelian (it is easy to show using the properties of trigonometric functions), while $O(2, \mathbb{R})$ is not because of the presence of "spatial inversions".

Linear transformations of orthonormal bases in $\mathbb{C}^2$: They can be expressed as matrices in the form

$$M = \begin{pmatrix} a_0 + ia_3 & a_2 + ia_1 \\ -a_2 + ia_1 & a_0 - ia_3 \end{pmatrix}, \text{ with } a_i \in \mathbb{R} \text{ s.t. } a_i a^i = 1$$

note that a_i satisfy the equation of a 3-sphere in $\mathbb{R}^4$; this can also be expressed as

$$M = \begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix}, \text{ with } |\alpha|^2 + |\beta|^2 = \det(M) = 1.$$

It is easy to show that matrices of this form are unitary:

$$MM^\dagger = M^\dagger M = \mathbb{I}.$$

It was shown before that

$$M = a_0 \mathbb{I} + ia_i \sigma^i + O(\mathbf{a}^2) :$$

this is equivalent to

$$M = \exp [i\varphi \hat{\mathbf{v}}_i \sigma^i]$$

where $0 \leq \varphi \leq \pi$ is an angle, $\hat{\mathbf{v}}$ is a unit vector in $\mathbb{R}^3$ and the exponential of a matrix is defined as

$$e^M := \sum_{n=0}^{\infty} \frac{M^n}{n!}.$$

Computing the exponential, it can be shown that these two expressions are equivalent: doing so is, in this case, quite easy using the properties of Pauli matrices

$$\begin{aligned} [\sigma^i, \sigma^j] &= 2i\varepsilon^{ij}_k \sigma^k \\ \{\sigma^i, \sigma^j\} &= 2\delta^{ij} \mathbb{I} \rightarrow (\sigma^i)^2 = \mathbb{I} \end{aligned}$$

from the second property follows that

$$\sigma^i \sigma^j = \frac{1}{2} (\sigma^i \sigma^j - \sigma^j \sigma^i + \sigma^i \sigma^j + \sigma^j \sigma^i) = \frac{1}{2} ([\sigma^i, \sigma^j] + \{\sigma^i, \sigma^j\})$$

From this and some calculations follows that

$$\exp [i\varphi \hat{v}_i \sigma^i] = \cos \varphi \mathbb{I} + i \sin \varphi \hat{v}_i \sigma^i.$$

Doing so explicitly for $\hat{v} = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$ yields

$$M = \sum_{n=0}^{\infty} \frac{(i\varphi \sigma^i)^n}{n!} = \sum_{k=0}^{\infty} \frac{(i\varphi \sigma^i)^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{(i\varphi \sigma^i)^{2k+1}}{(2k+1)!}$$

since $(\sigma^1)^{2k} = \mathbb{I}$ and $(\sigma^1)^{2k+1} = \sigma^1$, this becomes

$$M = \mathbb{I} \sum_{k=0}^{\infty} \frac{(-1)^k \varphi^{2k}}{(2k)!} + i\sigma^1 \sum_{k=0}^{\infty} \frac{(-1)^k \varphi^{2k+1}}{(2k+1)!} = \cos \varphi \mathbb{I} + i\sigma^1 \sin \varphi$$

8 Subgroups

Def. Given a group $(G, \cdot)$, a subset $H \subseteq G$ is a *subgroup* of G if and only if $(H, \cdot)$ is a group.

Observation: The trivial subgroups are $H = G$ and $H = \{e\} \subseteq G$; all other subgroups are called *proper* subgroups.

Def. A subgroup $H \subseteq G$ is called a *normal* subgroup if and only if

$$\forall h \in H, \forall g \in G, g \cdot h \cdot g^{-1} \in H.$$

Observation: If G is Abelian, every subgroup is normal.

Def. A group G is called a *semi-simple* group if and only if it does not admit a proper, normal and Abelian subgroup.

Def. A group G is called a *simple* group if and only if it does not admit a proper and normal subgroup.

Observation: All simple groups are semi-simple groups, but not vice-versa: a group that admits at least one proper, normal, non-Abelian subgroup and no Abelian subgroups is semi-simple but not simple!

9 Maps between groups

Def. A group *homomorphism* f is a map between two groups

$$\begin{aligned} f &: (G, \cdot) \rightarrow (Q, \circ) \\ f &: g \in G \rightarrow q \in Q \end{aligned}$$

such that

$$\forall g_1, g_2 \in G, f(g_1 \cdot g_2) = f(g_1) \circ f(g_2).$$

To define a group *isomorphism*, some other definitions are needed.

Def. Given a set $X \neq \emptyset$, a binary relation $\sim$ among its elements is an *equivalence relation* if and only if it is

- Reflexive, i.e. $\forall a \in X, a \sim a$;
- Symmetric, i.e. $\forall a, b \in X, a \sim b \rightarrow b \sim a$;
- Transitive, i.e. $\forall a, b, c \in X, a \sim b \wedge b \sim c \rightarrow a \sim c$.

Def. Given a set with an equivalence relation $(X, \sim)$, an *equivalence class* is a subset $Y \subseteq X$ such that

$$\forall a, b \in Y \subseteq X, a \sim b$$

Observation 1: All sets equipped with an equivalence relation have trivial equivalence classes, which are all the subsets containing only one element.

Observation 2: The equivalence classes of a set X must be disjoint: this follows from the definition of the equivalence relation.

Every equivalence class can be identified by one of its elements, called the *representative* of the class.

Def. A function $f : X \rightarrow Y$ is called

- *Injective* if and only if

$$\forall x_1, x_2 \in X, f(x_1) = f(x_2) \leftrightarrow x_1 = x_2;$$

- *Surjective* if and only if

$$\forall y \in Y, \exists x \in X \text{ s.t. } y = f(x);$$

- *Bijective* if and only if it is injective *and* surjective.

Def. A group *isomorphism* is a bijective group homomorphism.

Def. Two groups are *isomorphic* if $\exists$ an isomorphism f between them.

Theorem The property of isomorphism between two groups is an equivalence relation in the *set of all groups*.

Proof:

- The reflexive property is satisfied by the identity function, which is an isomorphism of a group in itself;
- $\forall G, Q$ groups, if $f : G \rightarrow Q$ is an isomorphism, the inverse $f^{-1} : Q \rightarrow G$ is an isomorphism;
- $\forall G, Q, L$ groups, if $f : G \rightarrow Q$ and $g : Q \rightarrow L$ are isomorphisms, then

$$g \circ f : G \xrightarrow{f} Q \xrightarrow{g} L$$

is an isomorphism, since the composition of group homomorphisms is an homomorphism and the composition of bijective functions is bijective. $\square$

10 The smallest finite groups

The smallest finite group is the *trivial group* and has order 1:

$$\mathbb{Z}_1 = \{e\}$$

its Cayley table is trivial

$$\begin{array}{c|c} \cdot & e \\ \hline e & e \end{array}$$

The second smallest group is the group of order 2

$$\mathbb{Z}_2 = \{e, a\}$$

and its Cayley table must be

$\cdot$	e	a
e	e	a
a	a	e

The group of order 3 is

$$\mathbb{Z}_3 = \{e, a, b\}$$

and its Cayley table is The missing entries can be found by brute force, i.e. trying all the 3 possible results

$\cdot$	e	a	b
e	e	a	b
a	a		
b	b		

and seeing if the result contradicts the assumption $a \neq b \neq e$ and/or the properties of the group: there is only one table that is self-consistent and it is

$\cdot$	e	a	b
e	e	a	b
a	a	$a^2 = b$	e
b	b	e	a

For bigger orders, there exist more than 1 self-consistent Cayley table, so how are cyclic groups defined?

Def. The cyclic groups $\mathbb{Z}_n$, with $n = 1, 2, \dots$ are in the form

$$\mathbb{Z}_n = \{e, a, a^2, a^3 \dots a^{n-1}\}$$

where $a^k = \underbrace{a \cdot a \cdot a \dots}_{k \text{ times}}$

They can be visualized as the set of the n 'th roots of 1 over the complex numbers $\mathbb{C}$:

$$\left\{ z_k \mid z_k = e^{i \frac{2\pi}{n} k}, \text{ with } k = 0, 1, \dots, n \right\}$$

with the correspondence $z_0 = e$, $z_k = a^{k-1}$